# PYTHON

## CONTROL STATEMENTS

Flow Control in programming refers to the ability to control the order in which statements and instructions are executed. It allows you to make decisions and repeat actions based on certain conditions. In python, flow control is achieved using conditional statements (if, elif, else) and loops (for, while).

Conditional statements allows you to executed different blocks of code based on certain conditions, while loops enable you to repeat a block of code until a specific condition is met.

### PYTHON INDENTATION:

In python, indentation plays a crucial role in defining the structure and scope of code blocks. It is used to group statements together and indicate which statement belong to a particular block of code. Python uses indentation instead of traditional braces or brackets of defining code blocks.

The standard convention in python is to use four spaces for indentation. It is recommended to be consistent with indentation throughout your code to ensure readability and maintainability.

Here's an example that demonstrate the use of indentation in python.

```
age = int(input("Enter age: "))
citizen = input("Are you a citizen (yes/no): ").lower() == "yes"

if age >= 18 and citizen:
    print("Eligible to vote")
else:
    print("Not eligible")
```

In the above example, the if statement is followed by an indented block of code, which is considered as the if-block.

### DECISION – MAKING STATEMENTS:

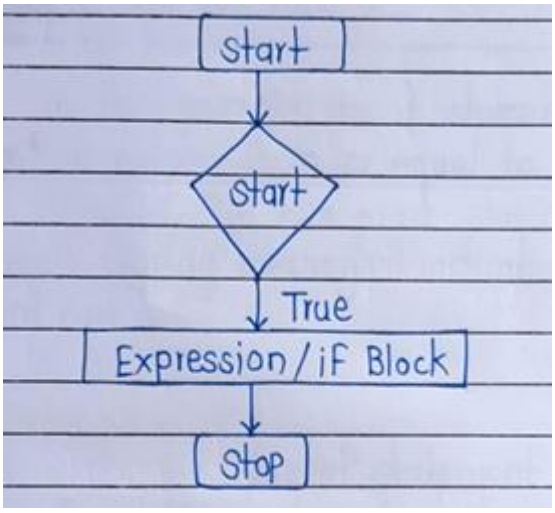
Decision-making statements is an important concept in programming, allowing us to execute specific block of code based on certain conditions. In python, we use different statements for decision-making.

- **If statement**: It tests a condition and executes a block of code if the condition is true.
- **If-else Statement**: It tests a condition and executes a block of code if the condition is true, and another block if the condition is false.
- **Nested-if statements**: It allows us to use an if-else statements inside another if statement, creating multiple levels of conditions and decisions.

1. **IF – STATEMENT:**

   The if – statement in python is used to perform a specific action or Execute a block of code if a condition is true. It allows us to make decisions in our programs based on the evaluation of the condition.

   FLOWCHART:




```
SYNTAX:
        If condition
                #code to be executed if the condition is true.
```

The condition is an expression that is evaluated to either True or False. If the condition is true, the code block indented below the if statement is executed. If the condition is false, the code block is skipped, and the program moves on the next statement after the if block.

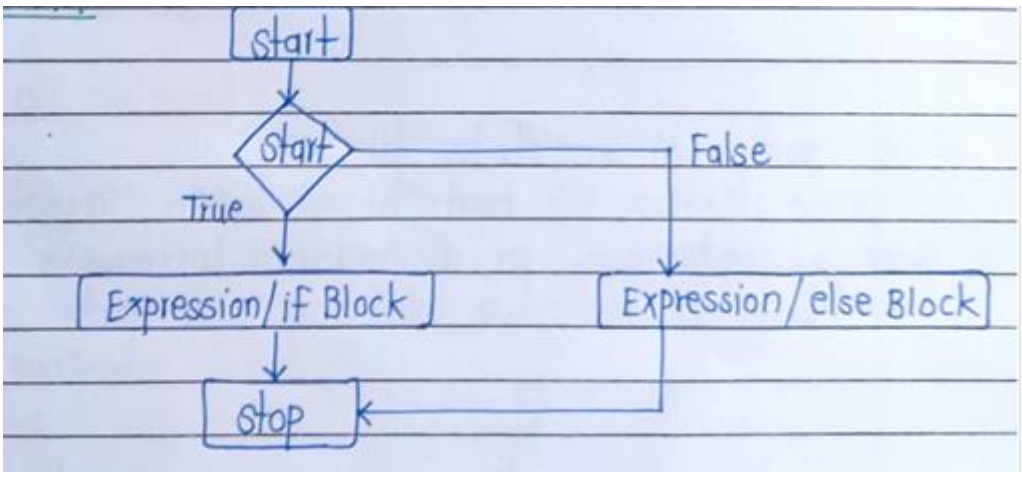
Eg.

```
a = 33
b = 200
if b > a:
  print("b is greater than a")
```

**2. IF – ELSE STATEMENT:**

The if-else statement in python provides a way to perform different actions based on the evaluation of a condition. It allows us to execute one block of code if the condition is true, and another block of code if the condition is false.

FLOWCHART:




SYNTAX:

```
If condition
        #code to be executed if condition is true
Else:
        #code to be executed if condition is false
```

The condition is an expression that is evaluated to either True of false, If the condition is true, the code block indented below the if statement is executed, If the condition is false, the code block indented below the else statement is executed.

Eg.

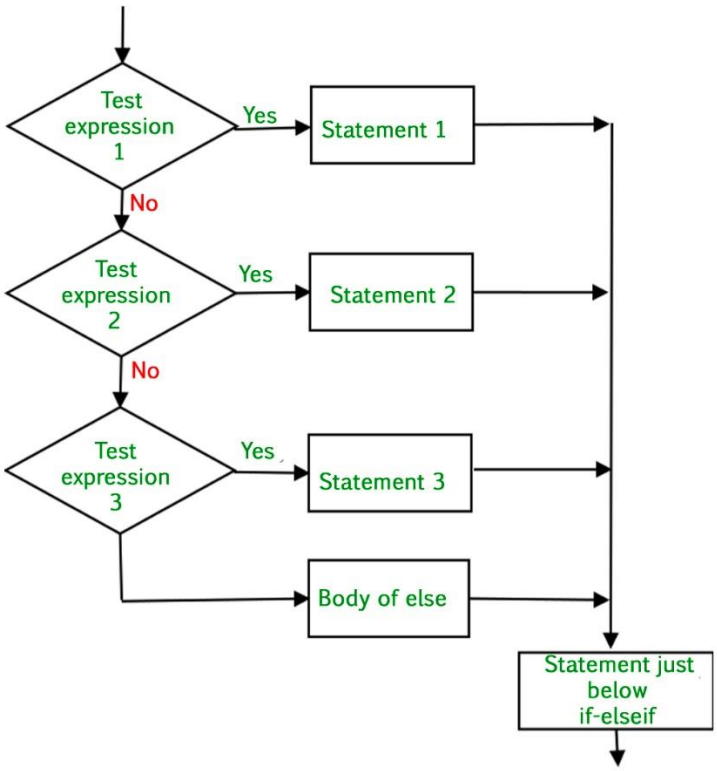
```
age = int(input("Enter age: "))
citizen = input("Are you a citizen (yes/no): ").lower() == "yes"

if age >= 18 and citizen:
    print("Eligible to vote")
else:
    print("Not eligible")
```

**3. ELIF STATEMENT:**

The elif statement in python is the short form for “else if”, allows us to check for multiple conditions in a sequential manner. It is used when we have more than two possible outcomes or conditions to evaluate.

FLOWCHART:



SYNTAX:

```
if condition1:
    # code to be executed if condition1 is true
elif condition2:
    # code to be executed if condition1 is false, condition2 is true.
elif condition3:
# code to executed if cond^n1, cond^n2 is false, condition3 is true.
else:
        # code to be executed if all conditions are false.
```

The conditions are evaluated in the order they are specified. If the first condition is true, the code block associated with that condition is executed, and the program skips the remaining conditions. If the condition is false, it moves on to the next condition and checks if it is true. This process continues until a condition is true, or if none of the conditions are true, the code within else statement is executed.

Eg.

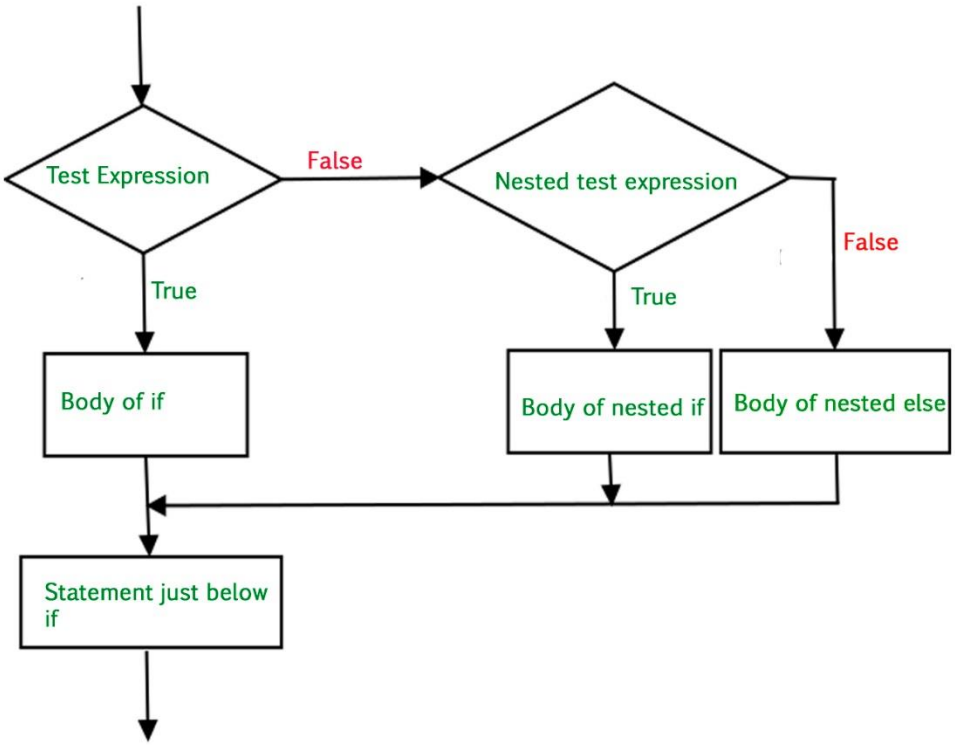
```
letter = "A"
if letter == "B":
    print("letter is B")
elif letter == "C":
    print("letter is C")
elif letter == "A":
    print("letter is A")
else:
    print("letter isn't A, B or C")
```

**4. NESTED IF STATEMENT:**

The nested if statement in python allows us to have an if-else statement inside another if statement. It enables us to create multiple levels of conditions and decisions in our code.

In a nested if statement, the inner if statement is indented under the outer if statement. The inner if statement is evaluated only if the condition of the outer if statement is true.

FLOWCHART:




SYNTAX:

```
if condition1:
    #code to be executed if condition1 is true
        if condition2:
    #code to be executed if condition1, condition2 is true
        else:
#code to be executed if cond^n1 is true but cond^n2 is false.
 else:
#code to be executed if condition1 is false
```

Eg,

```
age=25
income = 50000
if age>=18 :
  print ("You are eligible:")
  if income >= 30000:
    print (" You qualify for a loan.")
  else:
    print ("You do not qualify for a loan")
else:
  print(" You are not eligible:")
```

In the above example, the outer if statement checks if the age is greater than or equal to 18, if it is true the program executes the code within the outer if statement. Inside this code block, there is an inner if statement that checks if the income is greater than or equal to 30000.

## CONTROL STATEMENTS PROBLEMS

1. WAP to read num from user and do the following, if num is even then increase num by 10 else decrease num by 9.

2. WAP to read num from user and do the following, if num is greater than or equal to 100.0 then increase num by 25%, else decrease num by 30%.

3. WAP to read num from user and do the following, if the num greater than 100 then increase num by 10 else if num less than 100 then decrease num by 10 else increase by 40.

4. WAP to read salary from user and do the following if salary is greater than 5000.0 then increase salary by 20%, else if salary lesser than 5000.0 then decrease salary by 15%, else increase salary by 40%.

5. WAP to read a char input from user and do the following operation, if input is equal to ‘R’ then display ‘STOP’, else if input is equal to ‘Y’ then display ‘GET READY’, else if input is equal to ‘G’ then display ‘GO’ else display ‘invalid input’.

6. WAP to input from users to verify the given number is positive, if the number is positive, then check the number is even.

7. WAP to display to read attendance from the student and perform the below operation. If the student attendance is greater than 90 then read marks and do the following, if marks is greater than or equal to 80, then increase the marks by 5, else display “no grace marks”.

8. WAP to read Boolean input from the user and perform the below operation. If the input is true then read char input and do the following if the char input is ‘a’ then display “a for apple”, else if char input is ‘b’ then display “b for ball” else if the char input is ‘c’ then display “c for cat” else display “invalid char input” else display “no operation”.